1.7 Introductory organic chemistry

Related topics in Units 2, 4 and 5 will assume knowledge of this material.

Students will be assessed on their ability to:

1 Introduction

a demonstrate an understanding that there are series of organic compounds characterised by a general formula and one or more functional groups

b apply the rules of IUPAC nomenclature to compounds relevant to this specification and draw these compounds, as they are encountered in the specification, using structural, displayed and skeletal formulae

c appreciate the difference between hazard and risk

d demonstrate an understanding of the hazards associated with organic compounds and why it is necessary to carry out risk assessments when dealing with potentially hazardous materials. Suggest ways by which risks can be reduced and reactions can be carried out safely by:

- i working on a smaller scale
- ii taking specific precautions or using alternative techniques depending on the properties of the substances involved
- iii carrying out the reaction using an alternative method that involves less hazardous substances.

2 Alkanes

a state the general formula of alkanes and understand that they are saturated hydrocarbons which contain single bonds only

b explain the existence of structural isomers using alkanes (up to C_5) as examples

c know that alkanes are used as fuels and obtained from the fractional distillation, cracking and reformation of crude oil

d discuss the reasons for developing alternative fuels in terms of sustainability and reducing emissions, including the emission of CO_2 and its relationship to climate change

e describe the reactions of alkanes in terms of combustion and substitution by chlorine showing the mechanism of free radical substitution in terms of initiation, propagation and termination and using curly half-arrows in the mechanism to show the formation of free radicals in the initiation step using a single dot to represent the unpaired electron.

3 Alkenes

a state the general formula of alkenes and understand that they are unsaturated hydrocarbons with a carbon-carbon double bond which consists of a σ and a π bond

b explain E-Z isomerism (geometric/cis-trans isomerism) in terms of restricted rotation around a C=C double bond and the nature of the substituents on the carbon atoms

c demonstrate an understanding of the E-Z naming system and why it is necessary to use this when the *cis*- and *trans*- naming system breaks down

d describe the addition reactions of alkenes, limited to:

- i the addition of hydrogen with a nickel catalyst to form an alkane
- ii the addition of halogens to produce di-substituted halogenoalkanes
- iii the addition of hydrogen halides to produce mono-substituted halogenoalkanes
- iv oxidation of the double bond by potassium manganate(VII) to produce a diol

e describe the mechanism (including diagrams), giving evidence where possible, of:

- i the electrophilic addition of bromine and hydrogen bromide to ethene
- ii the electrophilic addition of hydrogen bromide to propene

f describe the test for the presence of C=C using bromine water and understand that the product is the addition of OH and Br

g describe the addition polymerisation of alkenes and identify the repeat unit given the monomer, and vice versa

h interpret given information about the uses of energy and resources over the life cycle of polymer products to show how the use of renewable resources, recycling and energy recovery can contribute to the more sustainable use of materials.